Module 2

Relations : Representation - matrix - graph types of relations, properties, equivalence relations, equivalence class.

Relations

In mathematics, we often deal with sets of numbers or objects and the ways they are connected. One of the important concept that help us describe this connection is relations.

- A relation is simply a connection between two sets of objects. Think of it as a rule that pairs elements from one set with the elements from another set.

- Simple Definition :
A relation is a set of ordered pairs that describe a relationship between elements.

Ordered Pair

An ordered pair consist of two elements written in a particular order.

It is written as:

$$(a, b)$$

Here,

a is the first element, b is the second elent.

- order is important.

$$(1, 2) \neq (2, 1)$$

- Two ordered pairs are equal if their corresponding elements are equal.

$$(a, b) = (c, d)$$

if and only if $a = c$ and $b = d$

- Example

$$(x, 4) = (2, 4)$$

then

$$x = 2 \text{ and } y = 4$$

Cartesian Product.

let $A = \{1, 2\}$ and $B = \{a, b\}$
The Cartesian Product of A & B is written as:
$A \times B$
It contains all possible ordered pairs where : the first element comes from A.
the second element comes from B.

So here, $A \times B = \{(1, a), (1, b), (2, a), (2, b)\}$

If set A contain m elements : $|A| = m$
set B contain n elements : $|B| = n$
Then
$A \times B$ has mn ordered pairs
$|A \times B| = mn$

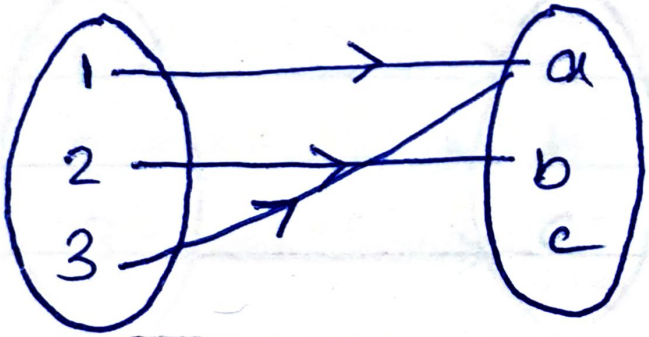
Example:
$|A| = 3$, $|B| = 4$
$|A \times B = 3 \times 4 = 12$

→ Let A and B be two sets.
A relation from A to B is any subset of A×B

$$R \subseteq A \times B$$

Example
let $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$
and $R = \{(1, a), (2, b), (3, a)\}$

Since $R \subseteq A \times B$, R is a relation from A to B



Domain: set of all first elements of the ordered pair in the relation
here, domain = {1, 2, 3}

Codomain: A Second set that was available to form the relation.
here, codomain = {a, b, c}

Range : set of all second elements of the ordered pair in the relation.
here, range = {a, b}

Representation of Relations

1. Ordered pair.
- In this set of ordered pairs of ~~a and~~ a and b are used to represent relation.

Example:
Let $A = \{1, 2, 3, 4, 5\}$ $B = \{2, 4, 9, 16, 25\}$
R is the relation of ordered pairs where the second number is the square of first number.

$$R = \{(a, b) : b = a^2\}$$

$$R = \{(1, 1), (2, 4), (3, 9), (4, 16), (5, 25)\}$$

This is the ordered pair representation of the above relation.

2. Matrix Representation

- This representation use a grid of zeros and ones for finite set.
- Rows represent elements from the first set
  Columns represent elements from the second set.
- Write 1 if the pair is in the relation and a 0 if it is not in the relation

Example

Let $A = \{1, 2, 3\}$ $B = \{0, 1, 2, 4\}$

and $R = \{(1,1), (1,2), (1,4), (2,2), (2,4), (3,4)\}$

Then the matrix representation of the relation

	0	1	2	4
1	0	1	1	1
2	0	0	1	1
3	0	0	0	1

→ elements from second set

↓ elements from first set

3. Graph Representation

→ A graph consist set of vertices and edges

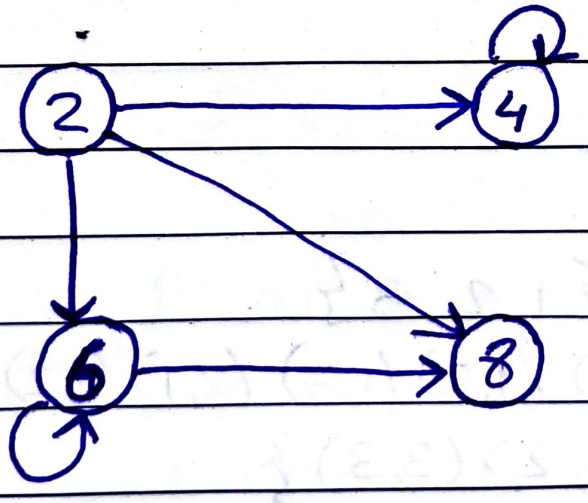
→ Each element of the set is represented by a point called nodes or vertices

→ An arrow is drawn from one node to its related node

Example:

Let $A = \{2, 4, 6\}$ and $B = \{4, 6, 8\}$

$R = \{(2,4)\ (2,6)\ (2,8)\ (4,4)\ (6,6)\ (6,8)\}$



This is the graph Representation of the above relation.

Types of Relation

1. Empty Relation : A relation R on a set A is called empty relation if $R = \phi$

Example:

If $A = \{3, 4\}$ and $R = \{(x, y) : x + y > 9\}$

Therefore $R = \phi$

2. Universal Relation : A relation R on a set A is called Universal Relation if $R = A \times A$

Example:

If $A = \{1, 2, 3\}$

Then $R = \{(1,1)\ (1,2)\ (1,3)\ (2,1)\ (2,2)\ (2,3)\ (3,1)\ (3,2)\ (3,3)\}$

3. Inverse Relation : Let R be a relation from set A to set B. Then the inverse of R is the relation from B to A. given by $R^{-1} = \{(y, x) : (x, y) \in R\}$

eg: If $R = \{(1,2), (3,2), (5,1)\}$
Then $R^{-1} = \{(2,1), (2,3), (1,5)\}$

4. Identity Relation: A relation R on a set A is called Inverse Relation if every element relates only to itself
$R = \{(x,x) : x \in A\}$

Example:
If $A = \{1,2\}$
$R = \{(1,1), (2,2)\}$

Properties of Relation

1. Reflexive Relation.
A relation R on set A is called reflexive if every element is related to itself.

$(a,a) \in R$ for all $a \in A$.

Example:

let $A = \{1, 2, 3\}$

and $R = \{(1,1)\ (2,2)\ (3,3)\ (1,2)\}$

Since $(1,1), (2,2), (3,3) \in R$

this relation is reflexive.

2. Irreflexive Relation

A relation is called irreflexive if no element is related to itself.

$(a,a) \notin R$ for all $a \in A$.

Example:

let $A = \{1,2,3\}$

and

$R = \{(1,2)\ (2,3)\ (3,1)\}$

There is no self pairs like $(1,1)\ (2,2)\ (3,3)$

Therefore R is irreflexive.

3. Symmetric Relation

A relation is called symmetric if $(a,b) \in R$ then $(b,a) \in R$

If a is related to b, then b must also be related to a.

Example

Consider a relation

$R = \{(1,2), (2,1), (2,3), (3,2)\}$

We have $(1,2) \in R$ and $(2,1) \in R$

Also $(2,3) \in R$ and $(3,2) \in R$

Therefore the relation is symmetric

4. Anti Symmetric Relation

A relation R on a set A is ant symmetric if :

$(a,b) \in R$ and $(b,a) \in R$ then $a = b$.

i.e If a is related to b and b is related to a, then a and b must be same.

Example:

let $A = \{1, 2, 3\}$

and $R = \{(1,1), (2,2), (3,3), (1,2)\}$

We have $(1,2) \in R$

but

$(2,1) \notin R$

There is no violation of antisymmetry

Hence, the relation is antisymmetric.

5. Transitive Relation

A relation is called transitive if $(a,b) \in R$, $(b,c) \in R$ then $(a,c) \in R$

If a is related to b, and b is related to c, then a must be related to c.

Example

$R = \{(1,2) (2,3) (1,3)\}$

We have

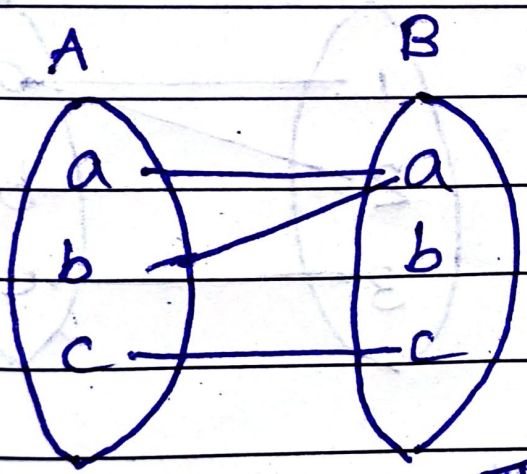
$(1,2) \in R$ $(2,3) \in R$ $(1,3) \in R$

Therefore,

the relation is Transitive

Functions

Let A and B be 2 non-empty sets. The relation which assign the element of A to the unique element of B is called function from A to B.
written as $f: A \to B$.

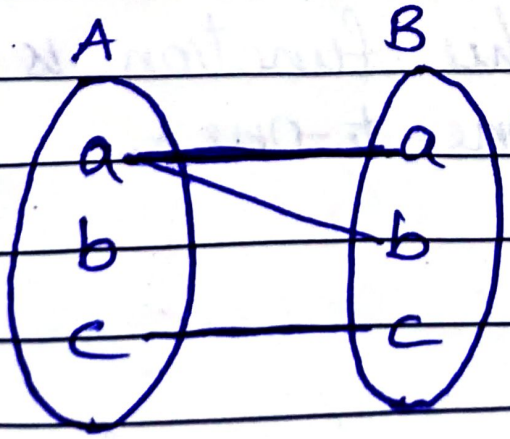


$f = \{(a,a), (b,a), (c,c)\}$

Domain $(f) = \{a, b, c\}$

Range $(f) = \{a, c\}$

This is an example of function.

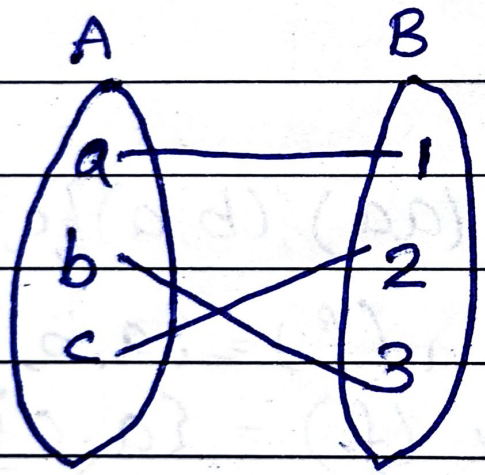


But this is not a function since element of A is assigned to two elements of B.

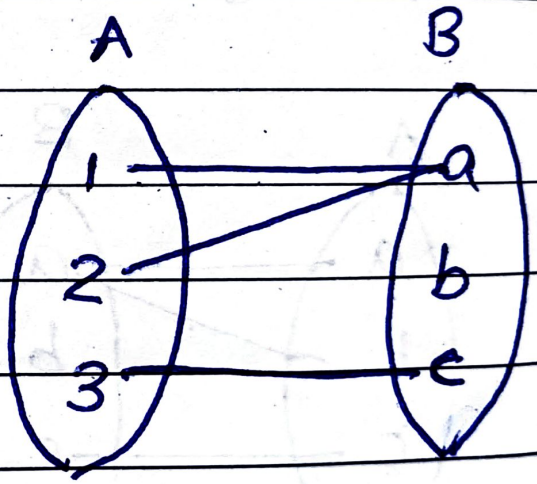
Types of function

1. One-to-One (1-1) or Injective

A function $f: A \to B$ is said to be 1-1 if different elements have distinct images.



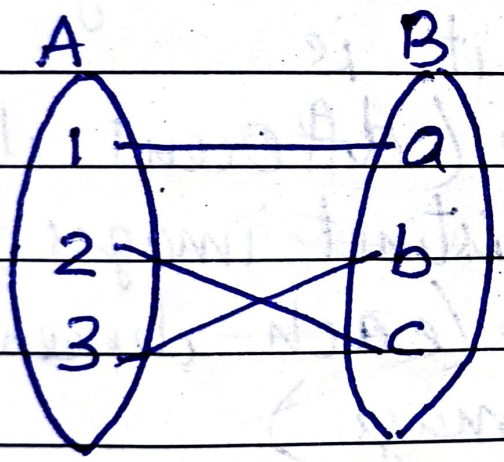
This is one-to-one function



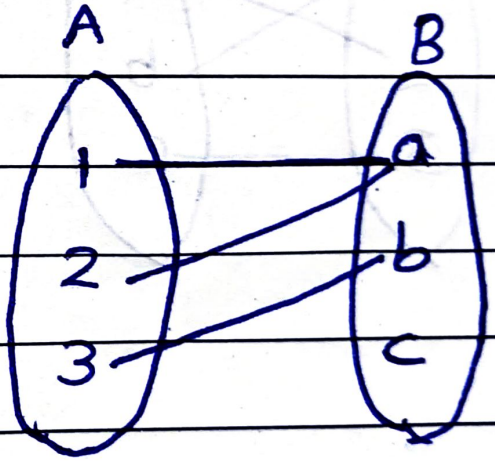
Here, a is the image of 2 elements. So this function is not one-to-one.

2. Onto or Surjective

A function $f: A \to B$ is said to be onto function if each element of B is the image of some element of A then A is onto.



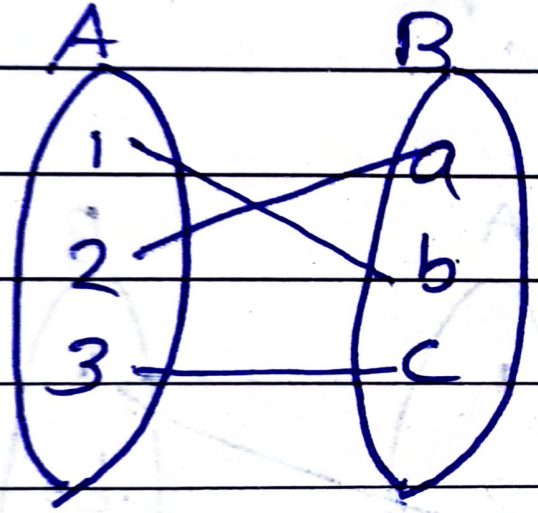
This is onto function



Here c is not assigned to any element of A So this function is not onto.

Bijective

A function $f: A \to B$ is said to be bijective if it is both one-one and onto.



This function is bijective since it is one-one (different elements have distinct images) and onto (each element in A has image)